

Sets

Informally, a set is a collection of "objects", which are called the elements of the set.

Examples for elements of a set:

numbers, vectors, matrices, strings, functions, other sets,
etc.

Describing sets

(a) list of elements inside "{" and "}"

Examples

$$A = \{2, 3, 5, 7\}$$

$$B = \{\text{blue, yellow, red}\}$$

$$C = \{\{1, 2\}, \{1, 3, 4\}, \{5\}\}$$

We usually use capital letters to denote sets.

This works only for (relatively small) finite sets.

(b) $D = \{1, 3, 5, 7, 9, 11, 13, \dots\}$ set of odd positive integers

$E = \{1, 2, 4, 8, 16, 32, 64, \dots\}$ powers of 2

(c) Set builder notation:

Define a set using a property.

The set consists of all elements that satisfy the property.

Examples

$$A = \{x \in \mathbb{R} \mid x^4 - 2x^2 - 3 \geq 0\}$$

$$B = \{n \in \mathbb{Z} \mid n = 2k \text{ for some } k \in \mathbb{Z}\} \text{ the set of even integers}$$

It is also common to write ":" instead of "|":

$$C = \{n \in \mathbb{N} : 2 \nmid n \text{ and } n \text{ is prime}\} \text{ all odd primes}$$

$$D = \{A \in M_{3 \times 3}(\mathbb{R}) : A = A^T\} \text{ all symmetric real } 3 \times 3 \text{ matrices}$$

Remark

- The order of elements is not significant

$\{2, 3\}$ is the same set as $\{3, 2\}$.

- no duplicates

$\{2, 3, 3\}$ is the same set as $\{2, 3\}$

Some common sets

$$\mathbb{N} = \{0, 1, 2, 3, \dots\}$$

$$\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$$

$\mathbb{Q}$ = rational numbers

$\mathbb{R}$ = real numbers

Notation: $\in$ and $\notin$

" $\in$ " stands for "is an element of"

" $\notin$ " stands for "is not an element of"

Examples

$$5 \in \mathbb{N}$$

$$-5 \notin \mathbb{N}$$

$$2/3 \in \mathbb{Q}$$

$$\sqrt{2} \notin \mathbb{Q}$$

Subsets

A is a subset of B if every element of A is also an element of B.

We write $A \subseteq B$.

If A is not a subset of B then we write $A \not\subseteq B$.

Examples

$$\mathbb{Q} \subseteq \mathbb{R}$$

$$\mathbb{Z} \not\subseteq \mathbb{N}$$

$$\{\sqrt{2}, \sqrt{3}, \pi\} \subseteq \mathbb{R}$$

Don't confuse $\in$ and $\subseteq$!

$$2 \in \mathbb{N} \checkmark \text{ but } \{2\} \subseteq \mathbb{N} \checkmark$$

$$\sqrt{3} \in \mathbb{R} \checkmark \text{ but } \{\sqrt{3}\} \subseteq \mathbb{R} \checkmark$$

$$\{\sqrt{2}, 5, 7, \pi\} \subseteq \mathbb{R} \checkmark$$

$$\{2\} \in \mathbb{N} \times$$

$$2 \subseteq \mathbb{N} \times$$

Equality of sets

Two sets are equal if they have the same elements.

Hence

$A = B$ if and only if $A \subseteq B$ and $B \subseteq A$.

Example If $A = \{1, 2, 3, 4\}$ and $B = \{4, 3, 2, 1\}$
then $A = B$.

The empty set

The empty set contains no element.

It is denoted by $\emptyset$, $\varnothing$, or $\{\}$.

For any set A , $\emptyset \subseteq A$.

Combining sets

The union of A and B , denoted $A \cup B$, contains exactly the elements appearing in A or B or both.

used in the
inclusive sense

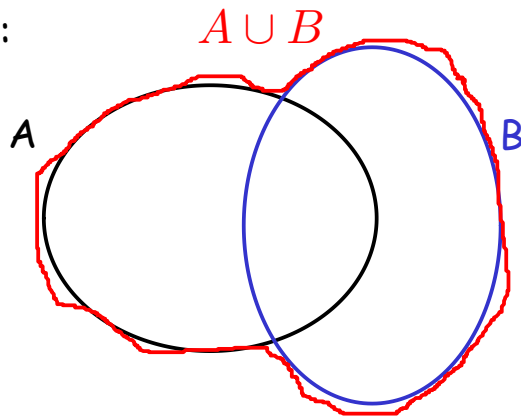
$$A \cup B = \{x \mid x \in A \text{ or } x \in B\}$$

Example

If $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6\}$

then $A \cup B = \{1, 2, 3, 4, 5, 6\}$.

Venn diagram:



The intersection of A and B , denoted $A \cap B$, contains exactly the elements appearing in both A and B .

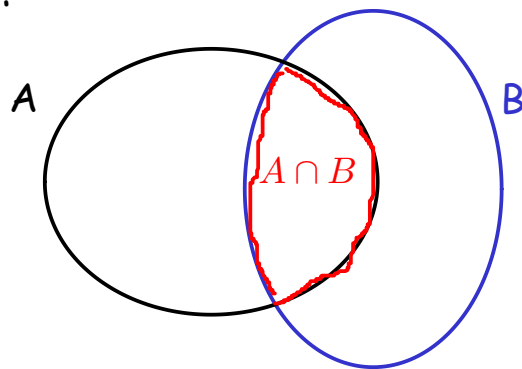
$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}$$

Example

If $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6\}$

then $A \cap B = \{3, 4\}$.

Venn diagram:



The set difference of A and B , denoted $A - B$, or $A \setminus B$ contains all elements that are in A but not in B .

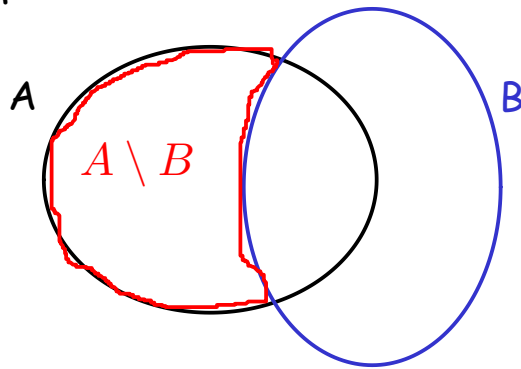
$$A \setminus B = \{x \mid x \in A \text{ and } x \notin B\}$$

Example

If $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6\}$

then $A \setminus B = \{1, 2\}$.

Venn diagram:



We say that two sets A and B are disjoint if $A \cap B = \emptyset$, i.e., if A and B don't have any elements in common.

Cartesian product

The cartesian product of A and B is the set $A \times B$, defined by

$$A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}.$$

We define

$$(a, b) = (c, d) \text{ if and only if } a = c \text{ and } b = d.$$

Example

1.) If $A = \{a, b, c\}$ and $B = \{a, d\}$

then $A \times B = \{(a, a), (a, d), (b, a), (b, d), (c, a), (c, d)\}$.

2.) $\mathbb{R} \times \mathbb{R} = \{(x, y) \mid x \in \mathbb{R}, y \in \mathbb{R}\} = \mathbb{R}^2$

